Dhirajzen Bagawath Geetha Kumaravel

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EDUCATION

New York University, MS in Computer Engineering | CGPA: 3.87/4.0 **Sept 2024 – May 2026**
Coursework: Generative AI Based Chip Design, SoC Design, Computing Systems Architecture, VLSI Systems & Architecture, Analog IC Design

College of Engineering Guindy, Anna University, BE Electronics and Communication Engineering | CGPA: 8.51/10 **Sept 2020 – May 2024**
Coursework: Digital System Design, Digital VLSI, Linear ICs, Operating Systems, Machine Learning, Advanced Microcontrollers

SKILLS

Languages: SystemVerilog, Verilog, Python, Tcl

Verification: UVM, Constrained-Random Verification, SystemVerilog Assertions (SVA), Functional Coverage

Protocols: AXI4, AXI4-Lite, APB, SPI, I2C

Tools: Synopsys VCS, Design Compiler, Cadence Xcelium, Genus, Innovus, Virtuoso, OpenLane, Xilinx Vivado

AI/ML for Verification: LangChain, LangGraph, RAG, ChromaDB, Prompt Engineering, Anthropic API

Hardware Concepts: RTL Design, Microarchitecture, Clock Domain Crossing (CDC), Static Timing Analysis (STA)

PROJECTS

LLM-Assisted Verification Assistant Using LangChain, LangGraph, and RAG

- Built a **RAG pipeline** over the 500-page ARM AXI4 specification using **LangChain**, **ChromaDB**, and Claude with regex preprocessing, semantic chunking, and **MMR retrieval** for grounded spec Q&A.
- Designed agentic coverage gap analyzer in **LangGraph** that parses functional coverage reports and suggests **UVM constrained-random stimulus** to close uncovered bins.

Verification of AXI4-Lite Slave Interface Using SystemVerilog and UVM

- Authored **Verification Plan** covering reset behavior, address alignment, and corner-case transactions.
- Built complete **UVM environment** (driver, monitor, sequencer, scoreboard) with constrained-random stimulus verifying handshake timing and back-to-back transaction stability.
- Achieved **functional coverage closure** validating protocol compliance across read and write channels.

Design Verification of APB Protocol Using SystemVerilog and UVM

- Authored **Verification Plan** covering reset behavior, PSLVERR responses, address alignment, and back-to-back transfers.
- Built **UVM environment** with scoreboard validating APB protocol compliance under constrained-random stimulus.
- Achieved **100% Functional Coverage** across address and data cross-bins during coverage closure.

Asynchronous FIFO for Robust Clock Domain Crossing (CDC)

- Verified **CDC-safe asynchronous FIFO** with independent read/write clock domains and Gray-code pointer synchronization.
- Implemented **SystemVerilog Assertions (SVA)** monitoring pointer stability and two-flop synchronizer latency across multiple clock ratios.
- Stress-tested under asynchronous reset, clock skew, and boundary conditions ensuring CDC robustness.

Functional Verification of SPI Protocol Using SystemVerilog and UVM

- Developed **UVM environment** for SPI controller with constrained-random read/write transactions and **functional coverage** tracking CPOL, CPHA modes.
- Designed **self-checking scoreboard** using 32×8 reference memory model validating readback data correctness.
- Verified **protocol compliance** during reset recovery and corner-case timing scenarios.

EXPERIENCE

New York University | New York, USA **June 2025 - May 2026**
Course Assistant | VLSI System & Architecture Design

- Automated grading pipeline using Python scripts to run student RTL submissions through VCS and generate structured reports.
- Prepared hands-on lab manuals and assignments for schematic/layout and digital design.